



This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

# SPRING AI

## GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

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### Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



## SPEECH-TO-TEXT

### Note

[illegible]

-  Simone Scannapieco

### Note

[illegible]

```

...
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
<dependency>
  <groupId>org.springframework.ai</groupId>
  <artifactId>spring-ai-starter-model-openai</artifactId>
</dependency>
...

```

```
spring:
  application:
    name: demo
  ai:
    openai:
      api-key: ${OPENAI_API_KEY}
```

### Note

[illegible]

```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final OpenAiAudioTranscriptionModel openAiAudioTranscriptionModel;

    AudioController(OpenAiAudioTranscriptionModel transcriptionModel) {
        this.openAiAudioTranscriptionModel = transcriptionModel;
    }

    @GetMapping("/transcribe")
    String transcribe(@Value("classpath:filename.mp3") Resource audioFile) {
        return openAiAudioTranscriptionModel.call(audioFile);
    }
}
```

### Note

[illegible]

- 1 Creazione interfaccia ed implementazione del servizio
- 2 Modifica controllore MVC
- 3 Test delle funzionalità con Postman/Insomnia

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Interfaccia servizio

```

package it.venis.ai.spring.demo.services;

import org.springframework.core.io.Resource;
import it.venis.ai.spring.demo.model.Answer;

public interface MultiModalityService {

    Answer getTranscriptionFromAudioFile(Resource audioFile);

}

```

Note

Note-taking area with horizontal lines.



```
package it.venis.ai.spring.demo.services;

...

@Service
public class MultiModalityServiceImpl implements MultiModalityService {

    private final ChatClient geminiChatClient;

    public MultiModalityServiceImpl(@Qualifier("geminiChatClient") ChatClient geminiChatClient) {

        this.geminiChatClient = geminiChatClient;

    }

    @Override
    public Answer getTranscriptionFromAudioFile(Resource audioFile) {

        String contentType = getContentType(audioFile);

        Media audioMedia = new Media(
            MimeTypeUtils.parseMimeType(contentType), audioFile);

        return new Answer(this.geminiChatClient.prompt()
            .user(u -> u.text("Trascrivi il seguente file audio: ").media(audioMedia))
            .call()
            .content());

    }

    ...
}
```

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```
...
private String getContentType(Resource multimediaFile) {
    String contentType = URLConnection.guessContentTypeFromName(multimediaFile.getFilename());

    if (contentType == null || contentType.equals("application/octet-stream")) {
        String fileName = multimediaFile.getFilename();
        if (fileName != null) {
            String extension = fileName.substring(fileName.lastIndexOf('.') + 1).toLowerCase();
            contentType = switch (extension) {
                case "mp3" -> "audio/mpeg";
                case "wav" -> "audio/wav";
                case "m4a" -> "audio/mp4";
                case "ogg" -> "audio/ogg";
                case "flac" -> "audio/flac";
                default -> "media/unknown";
            };
        } else {
            contentType = "media/unknown";
        }
    }

    return contentType;
}
}
```

### Note

[illegible]

```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@RequestMapping("/gemini/multi-modality")
public class GeminiMultimodalityController {

    private final MultiModalityService multiModalityService;

    public GeminiMultimodalityController(MultiModalityService multiModalityService) {

        this.multiModalityService = multiModalityService;
    }

    /**
     * Endpoint per la trascrizione di file audio in testo (Speech-to-Text).
     * Utilizza un file audio predefinito dal classpath e ne estrae la trascrizione testuale
     * tramite i servizi di riconoscimento vocale di Gemini.
     *
     * @param audioFile risorsa audio dal classpath (file WAV predefinito)
     * @return Answer oggetto contenente la trascrizione del file audio
     */
    @PostMapping("/stt")
    public Answer getTranscriptionFromAudioFile(@Value("classpath:Venis_descrizione_azienza.wav") Resource audioFile) {

        return this.multiModalityService.getTranscriptionFromAudioFile(audioFile);
    }
}
```

### Note

[illegible]

16-spring-ai-gemini-ollama-multimodality-audio-transcription

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